LinearLR#

https://pytorch.org/docs/stable/generated/torch.optim.lr_scheduler.LinearLR.

Description:

Decays the learning rate of each parameter group by linearly changing small multiplicative factor. The multiplication is done until the number of epoch reaches a pre-defined milestone: total_iters. Notice that such decay can happen simultaneously with other changes to the learning rate from outside this scheduler. When last_epoch=-1, sets initial lr as lr. optimizer (Optimizer) – Wrapped optimizer. start_factor (float) – The number we multiply learning rate in the first epoch. The multiplication factor changes towards end_factor in the following epochs. Default: 1./3. end_factor (float) – The number we multiply learning rate at the end of linear changing process. Default: 1.0.

Function Signature

```
class torch.optim.lr_scheduler.LinearLR(optimizer,
start_factor=0.3333333333333333, end_factor=1.0,
total_iters=5, last_epoch=-1)[source]#

Parameters

get_last_lr()[source]#

Return type

get_lr()[source]#
```

Returns:

dict[str, Any]

Code Examples

```
>>> # Assuming optimizer uses lr = 0.05 for all groups
>>> # lr = 0.003687  if epoch == 0
>>> # lr = 0.004875  if epoch == 1
>>> # lr = 0.006062  if epoch == 2
>>> # lr = 0.00725   if epoch == 3
>>> # ...
>>> # lr = 0.05      if epoch >= 40
>>> scheduler = LinearLR(optimizer, start_factor=0.05,
total_iters=40)
>>> for epoch in range(100):
>>>     train(...)
>>>     validate(...)
>>>     scheduler.step()
```

Detailed Documentation

Documentation

Decays the learning rate of each parameter group by linearly changing small multiplicative factor.

The multiplication is done until the number of epoch reaches a pre-defined milestone: total_iters. Notice that such decay can happen simultaneously with other changes to the learning rate from outside this scheduler. When last_epoch=-1, sets initial lr as lr.

optimizer (Optimizer) – Wrapped optimizer.

start_factor (float) – The number we multiply learning rate in the first epoch. The multiplication factor changes towards end_factor in the following epochs. Default: 1./3.

end_factor (float) – The number we multiply learning rate at the end of linear changing process. Default: 1.0.

total_iters (int) – The number of iterations that multiplicative factor reaches to 1. Default: 5.

last_epoch (int) – The index of the last epoch. Default: -1.

Return last computed learning rate by current scheduler.

Compute the learning rate.

Load the scheduler’s state.

state_dict (dict) – scheduler state. Should be an object returned from a call to state_dict().

Return the state of the scheduler as a dict.

It contains an entry for every variable in self.__dict__ which is not the optimizer.